SAY WATT ROBOTICS · FIELD LOG

Week 2.4 — The Data Makes It Home — SD Card to Radio to Laptop, and a CAD File That Got Overwritten

Sunday, October 11, 2026 · 3:00 – 5:00 PM · 12 students

Twelve of us at 2.4 — Maddox, Trenton, Adrian, Oliver, Eli, Liam, Rebecca, Annabelle, Rhylan, Frankie, Isaac, and Tyler — from 3:00 to 5:00. The headline is a pipeline. The ESP32 transmitter now stores its data on an SD card, reads it back, and sends it as one long string of text over the radio link. On the other end the string arrives at the laptop's serial port, and a Python script writes it out as a text file. Storage to radio to laptop to file: that is the data path the 2027 float needs, working end to end on a table, and it leads straight to the computer-generated graphs MATE scores at the station. Mechanical started printing the float's end cap and began the CAD for the buoyancy engine's syringe puller. Along the way the new model was saved over the original syringe-holder part — a lesson that costs exactly one afternoon, once. Electrical set out to lay out a printed circuit board in KiCad and came away with the first piece any board needs: the ESP32's pinout, drawn in KiCad. The manipulator trade got its scores updated, and the tie broke: linear actuator 32, all-hydraulic 25, the combination 20. The biggest swing was reliability — on prone-to-failure the actuator went from 3 to 5 while hydraulics fell from 4 to 2, which lines up with what the 2.1 leak test found about trapped air. Management built the official season budget spreadsheet, logged the newly purchased parts, started tracking refunds, and researched team registration for the events in Memphis and on Dauphin Island.

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STUDENTS AT THE SESSION	DATA PATH WORKING END TO END — SD CARD TO RADIO TO LAPTOP TO FILE	END CAP ON THE 3D PRINTER	LINEAR ACTUATOR OVER ALL-HYDRAULIC IN THE UPDATED MANIPULATOR TRADE

WHAT WE LEARNED

- Radio doesn't travel through water at these frequencies. A float measures underwater and talks at the surface, so it has to remember everything in between — that is the SD card's job.
- A data pipeline is only as good as its last step. A string on a serial port isn't data until something writes it down; the Python script is what turns a stream into a file you can graph.
- Never edit the only copy. Start a new part or save a new version before changing an old one — and know where your CAD tool keeps its history, because Onshape keeps every change and an overwritten part can usually be rolled back.
- A pinout is page one of every circuit board. You can't route a PCB until you know exactly which pin does what, and drawing the ESP32's pins in KiCad is that page.
- Track refunds like purchases. Money that comes back still has to land somewhere in the budget.
- Commands can go both ways. If the receiver can tell the float what to do after it is deployed, MATE wants that described in the float's documentation — so the design and the paperwork travel together.

DECISIONS MADE

- The float stores first and transmits second. Radio doesn't travel through water, so readings go to an SD card underwater and get sent as a text string once the float surfaces.
- The receiving end is a laptop running a Python script that catches the serial string and writes it to a text file — the starting point for the graphs MATE scores.
- The season has one official budget spreadsheet, with new parts logged as they arrive and refunds tracked in their own section.
- The float's first physical parts are coming off the printer while the buoyancy engine's syringe puller takes shape in CAD.
- The linear actuator leads the manipulator trade after the scores were updated — 32 to 25 over all-hydraulic, with the combination at 20.

ACTION ITEMS

- Finish the integrated master schedule
  Management · by 2.5
- Finish printing the end caps
  Mechanical group · by 2.5
- Finish the CAD models
  Mechanical group · by 2.5
- Finish soldering the buck converter
  Electrical group · by 2.5
- Find out whether the receiver can send control commands back to the profiler
  Software group · by 2.5
- Select a new marketing lead
  Management · by 2.5

STUDENT LEARNING OUTCOMES

- Built an end-to-end data path: ESP32 to SD card, SD card to radio, radio to laptop serial port, serial port to text file through a Python script.
- Started 3D printing the profiler's end cap.
- Began CAD of the buoyancy engine's syringe puller.
- Drew an ESP32 pinout in KiCad as groundwork for a custom circuit board.
- Updated the manipulator actuation trade study; the linear actuator now leads 32 to 25.
- Built the official season budget spreadsheet, logged new parts, and set up refund tracking.
- Researched team registration for the Memphis and Dauphin Island events.

LOOKING AHEAD

Finish the end caps and the CAD, finish the buck converter, and finish the integrated master schedule. On the software side the question flips direction — can the receiver send commands to the float, not just listen to it. And the marketing lead seat still needs filling.

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